Linux Security Basics Lab

This lab focuses on using Linux commands and understanding their output. Before starting, I downloaded the Ubuntu virtual machine and spent some time getting familiar with how it works. I completed the activity using the Ubuntu terminal to ensure safety regarding my own machine.

General System Tasks

1. The command to retrieve available updates is sudo apt update. Once enter is pressed, the system asks to authenticate sudo with a password. After the password is entered, available updates are shown.
   My system states that there are 64 packages that can be upgraded.

```
stuohey@Ubuntu:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://ports.ubuntu.com/ubuntu-ports questing InRelease
Get:2 http://ports.ubuntu.com/ubuntu-ports questing-updates InR
elease [136 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports questing-backports I
nRelease [133 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports questing-security In
Release [136 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports questing-updates/mai
n arm64 Packages [339 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports questing-updates/mai
n Translation-en [90.2 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports questing-updates/mai
n arm64 Components [36.6 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports questing-updates/mai
n arm64 c-n-f Metadata [7,264 B]
Get:9 http://ports.ubuntu.com/ubuntu-ports questing-updates/res
tricted arm64 Packages [241 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports questing-updates/re
stricted Translation-en [34.6 kB]
```

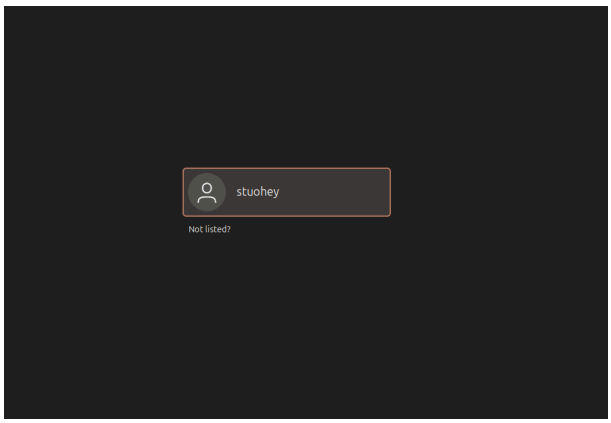
```
estricted Translation-en [34.0 kB]
Get:29 http://ports.ubuntu.com/ubuntu-ports questing-security/r
estricted arm64 Components [212 B]
Get:30 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse arm64 Packages [126 kB]
Get:31 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse Translation-en [45.7 kB]
Get:32 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse arm64 Components [10.8 kB]
Get:33 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse Icons (48x48) [8,563 B]
Get:34 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse Icons (64x64) [13.3 kB]
Get:35 http://ports.ubuntu.com/ubuntu-ports questing-security/u
niverse arm64 c-n-f Metadata [3,680 B]
Get:36 http://ports.ubuntu.com/ubuntu-ports questing-security/m
ultiverse arm64 Components [212 B]
Fetched 2,295 kB in 2s (1,121 kB/s)
64 packages can be upgraded. Run 'apt list --upgradable' to see
 them.
stuohey@Ubuntu:~$
```

2. Next, to actually install the updates, the command is sudo apt upgrade.

```
64 packages can be upgraded. Run 'apt list --upgradable' to see
 them.
stuohey@Ubuntu:~$ sudo apt upgrade
Upgrading:
  alsa-ucm-conf
  bpftool
  evolution-data-server
  evolution-data-server-common
  gir1.2-nm-1.0
  gir1.2-soup-3.0
  libasound2-data
  libasound2t64
  libatopology2t64
  libcamel-1.2-64t64
  libcurl3t64-gnutls
  libcurl4t64
  libebackend-1.2-11t64
  libebook-1.2-21t64
  libebook-contacts-1.2-4t64
  libecal-2.0-3
  libedata-book-1.2-27t64
  libedata-cal-2.0-2t64
```

3. To restart the server, the command is sudo reboot. Once the command is entered, it closes the virtual machine and reopens it, prompting the user to sign into Ubuntu again.

```
stuohey@Ubuntu:~$ sudo reboot
```



User Tasks

4. To change the current user to root, the command sudo su root can be used. This changes the prompt from the $ to a #. It also opened a new tab on the terminal for root.

```
stuohey@Ubuntu: ~        root@Ubuntu: /home/stuohey — sudo su root
stuohey@Ubuntu:~$ sudo su root
[sudo: authenticate] Password:
root@Ubuntu:/home/stuohey#
```

5. There are two ways to add new users – with the commands useradd and adduser. However, useradd simply adds the user without any password or further user information. Adduser, on the other hand, prompts you to create a password and add information for a home directory.

```
stuohey@Ubuntu:~$ sudo su root
[sudo: authenticate] Password:
root@Ubuntu:/home/stuohey# sudo useradd bobby
root@Ubuntu:/home/stuohey# sudo adduser sally
New password:
BAD PASSWORD: The password is shorter than 8 character
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password contains the user name in s
form
Retype new password:
passwd: password updated successfully
Changing the user information for sally
Enter the new value, or press ENTER for the default
        Full Name []: Sally
        Room Number []: 1
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
root@Ubuntu:/home/stuohey#
```

6. To change the current user to the one I previously created, Sally, the command is sudo su sally. Because Sally is not the root user, the prompt has a $ again.

```
root@Ubuntu:/home/stuohey# sudo su sally
sally@Ubuntu:/home/stuohey$
```

7. When Sally tries to add a new user with the name Earl, the command line returns an error that doesn't allow this. This is because Sally is a regular user and does not have root privileges. To allow Sally to add users, I could add her to the sudo group by the command usermod -aG sudo sally as root, then as sally I would be able to run sudo useradd earl successfully.

```
sally@Ubuntu:/home/stuohey$ sudo useradd earl
sudo-rs: I'm sorry sally. I'm afraid I can't do that
sally@Ubuntu:/home/stuohey$
```

8. To exit out from Sally's account and back to my own, the command is exit until I am back into my own account. Once there, I can delete a user bobby by the command sudo userdel bobby.

```
sally@Ubuntu:/home/stuohey$ exit
exit
root@Ubuntu:/home/stuohey# exit
exit
stuohey@Ubuntu:~$ sudo userdel bobby
[sudo: authenticate] Password:
stuohey@Ubuntu:~$
```

9. To change the password of Sally, the command sudo passwd sally is used. If a password with low security is entered, the system will print out the reason why it is a bad password – however still lets you use that password.

```
stuohey@Ubuntu:~$ sudo passwd sally
New password:
BAD PASSWORD: The password contains the user name in s
form
Retype new password:
passwd: password updated successfully
stuohey@Ubuntu:~$
```

10. Although it is easier to complete tasks and commands, it is bad practice to stay logged in as root. Without the need to type sudo and enter a password, it is much easier to make mistakes or typos that could be extremely detrimental to the system. If someone malicious gains access while a user is logged into root, they now would have full access to the entire system–including the ability to add and delete files, users, and install malware. Finally, when using root directly, there is no accountability as the system cannot track which specific user performed actions.
11. The command to see the user ID, or uid, is simply id.

```
stuohey@Ubuntu:~$ id
uid=1000(stuohey) gid=1000(stuohey) groups=1000(stuohey),2
7(sudo)
stuohey@Ubuntu:~$
```

Group Tasks

12. To display what groups the user belongs to, use the command groups.

```
stuohey@Ubuntu:~$ groups
stuohey sudo
```

13. To add sally to the sudo group, the command sudo usermod -aG sudo sally is used. Then, to have sally add a user named billy, the command is sudo useradd billy.

```
stuohey@Ubuntu:~$ sudo usermod -aG sudo sally
stuohey@Ubuntu:~$ sudo su sally
```

```
sally@Ubuntu:/home/stuohey$ sudo useradd billy
[sudo: authenticate] Password:
```

14. To create a new group, use the command sudo groupadd. I created a new group called cybersec.

```
stuohey@Ubuntu:~$ sudo groupadd cybersec
stuohey@Ubuntu:~$
```

15. To add sally to the cybersec group, use the command sudo usermod -aG cybersec.

```
stuohey@Ubuntu:~$ sudo usermod -aG cybersec sally
```

16. To check what groups sally is in, the command groups sally is used. This can also be shown by using id sally which prints out sally's uid, gid, and all of sally's groups. Additionally, if sudo su sally was used, then sally's groups could be printed with simply either groups or id.

```
stuohey@Ubuntu:~$ groups sally
sally : sally sudo users cybersec
stuohey@Ubuntu:~$ id sally
uid=1002(sally) gid=1002(sally) groups=1002(sally),27(sudo
),100(users),1004(cybersec)
```

Permission and Access Control Lists

17. To create a new directory, the mkdir command is used. To see the permissions, you can use the command ls -ld lab1. As shown in the image, the owner is stuohey and the group owner is stuohey. The letters at the start define permissions for the owner, for the group, and for anyone else. Owner: rwx (read, write, execute), Group: rwx (read, write, execute) Other: r-x (read, execute).

```
stuohey@Ubuntu:~$ mkdir lab1
stuohey@Ubuntu:~$ ls -l lab1
total 0
stuohey@Ubuntu:~$ ls -ld lab1
drwxrwxr-x 2 stuohey stuohey 4096 Feb 27 20:34 lab1
```

18. I changed to the lab1 directory by using the command cd lab1. Then, I created a new file using the touch command and named it helloWorld. I wrote a C program as shown to print out "Hello World!" on the terminal.

```
stuohey@Ubuntu:~/lab1$ touch helloWorld.c
stuohey@Ubuntu:~/lab1$ nano helloWorld.c
stuohey@Ubuntu:~/lab1$ gcc helloWorld.c -o helloWorld
stuohey@Ubuntu:~/lab1$ ./hello
bash: ./hello: No such file or directory
stuohey@Ubuntu:~/lab1$ ./helloWorld
Hello World!
stuohey@Ubuntu:~/lab1$
```

```
  GNU nano 8.4          helloWorld.c *
#include <stdio.h>

int main(){
        printf("Hello World!\n");
        return 0;
}
```

19. When entering the command ls -l helloWorld, it is shown that the permissions for the owner are reading, writing, and executing; the permissions for the group are reading, writing, and executing; and the permissions for other are reading and executing.
    a. The file already has w and x permissions for the group.

```
stuohey@Ubuntu:~/lab1$ ls -l helloWorld
-rwxrwxr-x 1 stuohey stuohey 70416 Feb 27 23:16 helloWorld
```

20. To view the Access Control List, or ACL, of the file, you can use the command getfacl helloWorld. This shows not only user, group, and other; but also would show any additional users with access to this file.

```
stuohey@Ubuntu:~/lab1$ getfacl helloWorld
# file: helloWorld
# owner: stuohey
# group: stuohey
user::rwx
group::rwx
other::r-x
```

21. To add to a user's permissions to a file, you can use the setfacl command. This line is saying that I am modifying the ACL to include user sally to have read and write permissions for the file helloWorld. When getfacl is ran again, sally has been added!

```
stuohey@Ubuntu:~/lab1$ setfacl -m u:sally:rw helloWorld
stuohey@Ubuntu:~/lab1$ getfacl helloWorld
# file: helloWorld
# owner: stuohey
# group: stuohey
user::rwx
user:sally:rw-
group::rwx
mask::rwx
other::r-x
```